WEEK 5: ASSIGNMENT

1. If a three-phase AC-DC converter operating with unity power factor, then to obtain the power flow from AC to DC, the voltage phasor obtained from the converter must

   A. lead the source voltage phasor
   B. lag the source voltage phasor
   C. align along the source voltage phasor
   D. be at an angle of 90 degree from the source voltage phasor

Answer: B

2. A 22kW, 3-phase, AC-DC converter fed from 400V, 50Hz mains. It is operated to generate 750V DC output voltage. If the designer kept the carrier wave which is varying between +3/-3V and the positive peak value of modulation wave is 2.7V. Then calculate the per phase inductance (in mH) needed to be kept. (Note: The answer should be rounded up to 3 decimal places and use $\pi = 3.14$). Assume converter is lossless.

Answer: 5 - 7

3. If a three-phase PFC converter, switching at 5 kHz with perfect unity power factor current drawn is fed from 400V, 50Hz AC input, then output capacitor voltage ripple has following dominant frequency component (having $V_m < V_c$)
   A. 50 Hz
   B. 100 Hz
   C. 5 kHz
   D. 300 Hz

Answer: C

4. A half-bridge voltage source converter outputs a sine PWM signal with V_{dc} DC link. What will be RMS value of the component of fundamental component in the output phase voltage for modulation index m.

A. mV_{dc}

B. $\frac{mV_{dc}}{2}$

C. $\frac{mV_{dc}}{\sqrt{2}}$

D. $\frac{mV_{dc}}{2\sqrt{2}}$

Answer: D

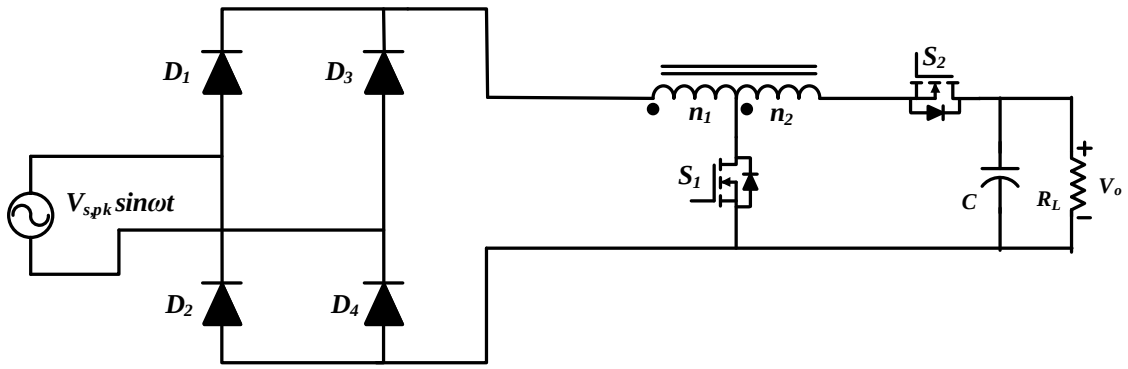
5. If the half-bridge based voltage source converter is operated using sinusoidal PWM, the carrier frequency is 10kHz, while the fundamental frequency is 50Hz, then which of the following harmonic frequency exists in the output voltage

A. 10050
B. 10150
C. 20100
D. 10100

Answer: A, B, C

6. For the PFC incorporating following DC-DC converter as shown below, What will be the expression of duty ratio (d(t)). Symbols have obvious meanings.

Note: The switches S_1 and S_2 are switched in a complementary manner with S_1 is in conduction during $d(t)T_s$, while S_2 is in conduction for $(1\text{-}d(t))T_s$.



A. $1 - \frac{\frac{n_2}{n_1}|v_s(t)|}{V_o}$

B. $\frac{V_o - |v_s(t)|}{V_o + \frac{n_2}{n_1}|v_s(t)|}$

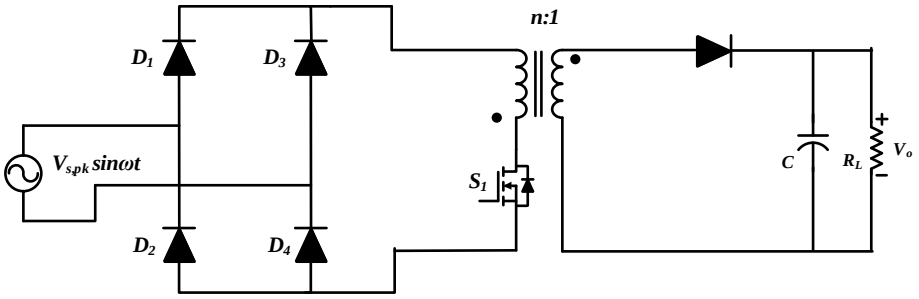
C. $\frac{V_o + |v_s(t)|}{V_o - \frac{n_2}{n_1}|v_s(t)|}$

D. $\frac{1}{1 + \frac{\frac{n_2}{n_1}|v_s(t)|}{V_o}}$

Answer: B

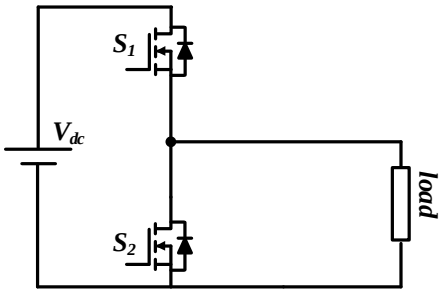
7. For the PFC incorporating Flyback DC-DC converter as shown below is fed from 230V, 50Hz AC voltage. If the duty ratio of the switch S_1 at the positive peak of the input AC voltage is kept at 0.12, then what will be the output voltage, if $n = 0.6$.

Assume the switches are ideal and the converter is lossless. The answer should be rounded up to 2 decimal places.



Answer: 70-75

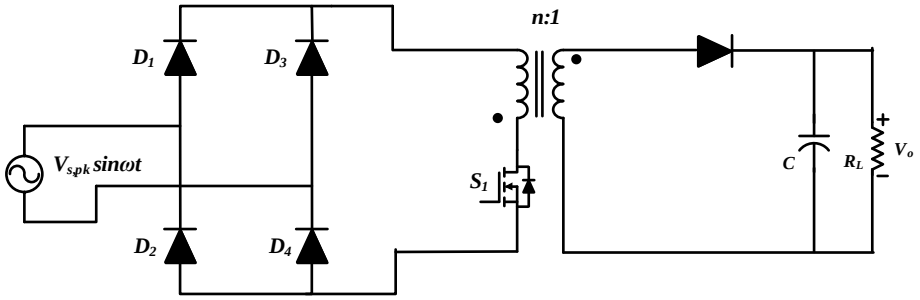
8. A half-bridge voltage source converter as shown below operates under PWM with 200V DC link (V_{dc}). What is the average output voltage, if the gate pulses are generated by comparing the carrier and modulating signal with a condition that if modulating wave > carrier wave, Top switch (S_1) is ON applying V_{dc} across the load, else the bottom switch (S_2) is ON. The carrier used is the triangular carrier which goes between 5 V to 0 V and the modulating signal is maintained at the 4 V value. The answer should be rounded to nearest integer. Symbols have the obvious meaning.



Answer: 158-162

9. For the PFC incorporating Flyback DC-DC converter as shown below is fed from 240V, 50Hz AC voltage. What will be the duty ratio at the positive peak of the input AC voltage, when the output DC link voltage is kept at 400V and n = 2 .

Assume the switches are ideal and the converter is lossless. The answer should be rounded up to 2 decimal places.



Answer: 0.64 to 0.67

Answer Key:

1. B	2. 5 to 7
3. C	4. D
5. A, B, C	6. B
7. 70 to 75	8. 158 to 162
9. 0.64 to 0.67	